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Abstract

This project addresses the recent increase of interest of artificial intelligence in the modern healthcare system, specifically the need for more accurate and robust systems in the care of type one diabetes. Moreover, the challenge of maintaining healthy blood sugar levels is controlled by many factors, including manual insulin boluses that a given diabetic individual would need to perform multiple times daily. This provides an opportunity for machine learning techniques to individually model a diabetic subject's blood sugar and provide a more accurate bolus amount. This project started as a simple NLP project using Nutrionix API tool to return macronutrients of a user-inputted meal. However, incorporating insulin bolus calculations was a natural step forward and a relevant application of the initial code.

1 Introduction

Type 1 diabetes mellitus, or more commonly known as type 1 diabetes, is a chronic autoimmune disorder in which the pancreas produces little to no insulin. With no cure, the patient must self-inject insulin to maintain healthy blood sugar levels. This occurs throughout the day, most often after meals. To properly manage levels, it is essential to incorporate carbohydrate intake, current blood glucose, and activity level while calculating an appropriate insulin bolus[1]. These represent only a few of the many variables that must be taken into account in boluses. The complexity in itself fully justifies the use of a device that facilitates bolus calculation and the need for bolus calculators (BCs) is further substantiated by studies documenting a high frequency of poor numeracy skills among adults with type one diabetes (T1D). Despite the fact that the exact amount of carbohydrates in a meal was given to 201 adult T1D patients, 57% came to a false result by manual insulin dose calculation. However, when provided with an automated BC the proportion of false results was reduced to 7%. [2]Furthermore, the addition of artificial intelligence into such a system would support patients in an otherwise stressful and resource demanding calculations, with the potential to incorporate past blood sugar trends of a patient.

Introducing Diabot: a prototype of an all-in-one program in which one can input a meal and receive a personalized meal insulin dosage. The prototype consists of a basic text processing model to extract food items and the corresponding amounts. It then returns a dosage in units of insulin. It is important to note that a unit is not an objective measurement; this varies patient to patient, and Diabot assists with this via arguments passed through the command line. This will be discussed in detail in the documentation section. Diabot's allows personalized dosage recommendations through CGM (Continuous Glucose Monitor) Data. CGMs are devices that measure blood glucose levels every five minutes to help a patient stay in a healthy blood sugar range. This data is especially helpful, as Diabot then finds simple averages in 4 time periods throughout the day and increases or decreases the bolus. This will also be explained in more detail later. For reference, my 3-month CGM data is included with the code to illustrate the effect on insulin bolus calculations.

2 Nutrionix

To provide accurate bolus amounts, it is crucial to have a large database of all foods that contains information of carbohydrates, sugar, protein, and more. Diabot uses the Nutrionix API, which is called with a certain amount and a food item. A more detailed list of measurement units accepted will be provided in the documentation. Diabot returns all macronutrient information, even though it doesn't account for all of them in the bolus calcula-

[1] Davidson et al. (2008)

[2] Davidson et al. (2008)

tion, as carbohydrates remain the most important in calculating insulin dosages[3].

2.1 Text Processing of User Input

Diabot uses spaCy to tokenize and produce a document with part-of-speech tagging. It then finds a quantity or numeric token, and looks at the next three tokens to attempt to find a measurement unit. Once a measurement unit is found, it then looks for a noun chunk, which is assumed to be a food item. The processing of inputs is supposed to be user-friendly, allowing the patient to input sentences in a natural way, for example: "I drank two cups of milk", or "I ate three tablespoons of peanut butter". These examples are accepted by Diabot. It should be noted that simple sentences work best, and multiple food items with varying quantities can be passed as long as it follows a numeric-unit-food structure.

3 Bolus calculation

A bolus is defined as giving a patient a dose of insulin. To properly bolus a meal, a popular method is utilized called 'carb-counting'. The concept revolves around the fact that carbohydrates account for the majority of the bolus calculation. Hence, one half of the bolus calculation is simply the amount of carbohydrates in a meal divided by a patient's ICR (Insulin to carbohydrate ratio). Because this is a hyperparameter that varies from patient to patient, this is one of the arguments that Diabot needs to calculate a bolus. Another aspect of the calculation is the patient's target blood sugar. This also varies patient to patient, hence it is also a parameter. Diabot gives the option to input the patient's current blood sugar and target blood sugar, which would then calculate the correction portion of the bolus.

$$\text{Bolus}_{\text{insulin}} = \text{Meal}_{\text{insulin}} + \text{Correction}_{\text{insulin}}$$

This is the general equation for calculating a bolus. The formal equation is as follows:

$$\text{Bolus}_{\text{insulin}} = \frac{\text{CHO}}{\text{ICR}} + \frac{\text{CurrentBG} - \text{TargetBG}}{\text{CF}}$$

Where CHO is the total carbohydrate count in grams and ICR is the patient's insulin-to-carb ratio, or formally, it is the amount of CHO needed to match the BG lowering effect of 1 unit of rapid-acting insulin[4]. CurrentBG is simply the patient's current blood sugar, while TargetBG is the target blood glucose level. These two values are divided by a patient's CF (correction factor). Correction factor is another parameter, and is defined as the amount of CHO needed to match the BG lowering effect of 1 unit of rapid-acting insulin[5].It should be noted that Diabot does not allow recommendations of boluses above 10 units as a safety precaution.

3.1 Physiologic State

As defined by Davidson et al, often, the sum of the meal and correction insulin part should be multiplied by a proportion reflecting the current physiologic state (PS) of the patient. When insulin sensitivity is increased, for example, due to physical activity, the PS is < 1 and when insulin sensitivity is decreased, for example, during illness, the PS is > 1. The PS is subject to large intra- and inter-individual variation and since there is no research supporting quantification of the proportion, the patient is left to a trial-and-error approach.
Moreover, insulin has a long-lasting effect. This introduces another parameter: IOB (Insulin on Board). This is an essential part of the calculation, as insulin typically lasts more than 5 hours with today's rapid-acting analogs. One study found that 64% of all pump boluses are given less than 4.5 hours after the previous bolus. Another study found that the average number of carbs and correction boluses was 4.14 and 2.12 per day, respectively, during a 10-week period of pump wear among 396 pump wearers[6]. Not accounting for IOB leads to 'insulin stacking', which increases the chance of hypoglycemic events. With this approach we update our bolus equation to the following:

$$\text{Bolus}_{\text{insulin}} = \frac{\text{CHO}}{\text{ICR}} + \frac{\text{CurrentBG} - \text{TargetBG}}{\text{CF}} \times \text{PS} - \text{IOB}$$

Diabot takes a different approach to PS than that of measuring physical activity due to this variance. In a sense, 'measuring' physical activity is essentially impossible to reproduce without a large scale of personal data and hence personal habits. Diabot solves this with a more data-driven analysis, which is performed through CGM data to scale the final bolus calculation. If CGM data is not provided, this factor isn't included in the final bolus calculation.

[3] Schmidt et al. (2014)

[4] Davidson et al. (2008)

[5] Davidson et al. (2008)

[6] Walsh et al. (2011)

First, CGM data is re-indexed to hourly based readings. Then, averages are calculated for four time periods:

Time of Day	**Average Glucose (mg/dL)**
Morning (8am–12pm)	131.8
Afternoon (12pm–6pm)	163.4
Evening (6pm–10pm)	158.2
Night (10pm–8am)	160.1

As depicted in the table, the four periods are not equal in length; instead they serve as a proxy for time periods around meals. It is also noteworthy that the morning average is much lower than the afternoon, evening, and night. This illustrates that trends in blood sugar can be used to adjust bolus amounts in a meaningful way.
Through these averages, Diabot calculates TBF, or Temporal Bolus Factor. To illustrate this,

Let:

- G_h = average glucose for hour h
- $\overline{G}$ = overall average glucose

Then we calculate our relative factor:

$$R_h = \frac{G_h}{\overline{G}}$$

It is up to debate the extent of effect that PS, or in this context TBF, should have on the final bolus calculation. It seems reasonable to scale in the range of the highest average glucose divided by the lowest average glucose. To simplify the visualization:

$$\text{Sens}_{min} = \frac{\text{LowestAvgBG}}{\text{HighestAvgBG}}$$

and

$$\text{Sens}_{max} = \frac{\text{HighestAvgBG}}{\text{LowestAvgBG}}$$

The final calculation for TBF is then:

$$TBF = \max\left(\text{Sens}_{\text{min}}, \min\left(\text{Sens}_{\text{max}}, 2 - R_h\right)\right)$$

To apply the TBF to the final bolus, it is multiplied to both the ICR and CF:

$$ICR' = ICR \cdot TBF$$
$$CF' = CF \cdot TBF$$

Finally, the meal bolus is now:

$$\text{Carb Bolus} = \frac{\text{CHO}}{ICR'}$$

It can be seen that this has the desired effect, as for a $TBF' > 1$, the meal bolus is lowered, and for $TBF < 1$, the meal bolus is increased. Furthermore, if CurrentBG and TargetBG are given, the Correction Bolus is also modified:

$$\text{Correction Bolus} = \frac{\text{CurrentBG} - \text{TargetBG}}{CF'}$$

And the final bolus calculation is as follows:

$$\text{Final Bolus} = \text{Carb Bolus} + \text{Correction Bolus} - \text{IOB}$$

Note that this final bolus has a maximum value of 10 units for safety.

4 Diabot Documentation

After downloading the requirements, to run the Python code, the user can use the following command-line arguments:

Argument
`-bg`
`-target`
`-icr`
`-cf`
`-iob`
`-cgm-data`

Where

- `-bg`: Your current blood glucose level
- `-target`: Your target blood glucose level (Default: 100)
- `-icr`: Your insulin-to-carb ratio (Default: 1:15)
- `-cf`: Your correction factor (Default: 1:50)
- `-iob`: The amount of insulin on board (Default: 0)
- `-cgm-data`: Path to a file with continuous glucose monitor (CGM) data

As well as numeric values, Diabot accepts the following as measurement units: "one", "gram", "grams", "g", "ounce", "ounces", "oz", "pound", "pounds", "lb", "lbs", "cup", "cups", "tbsp", "tablespoon", "tablespoons", "tsp", "teaspoon", "teaspoons"

Limitations

The limitations of Diabot lies in its inability to correctly account for the myriad of variables in calculating boluses. Although time-of-day adjustments are made, this doesn't truly reflect physiological state. Furthermore, fat and protein are also macronutrients that have an effect on blood sugar. Diabot does not take these into account, which does limit it's accuracy. Although these calculations are not complex, it remains true that carbohydrate is the nutrition component with the greatest impact on postprandial blood glucose level and that the effect of protein and fat is negligible and covered by the basal insulin[7]. Basal insulin, for reference, is non self-administered insulin. It is most often administered automatically through diabetic pumps. Diabot is a prototype, and unfortunately bolus data wasn't available from Dexcom or Tandem. This is why a more complex model wasn't used in the code, as there wasn't lots of data to operate on. If bolus data and a food log were recorded, alongside an activity monitor, the potential for a machine learning model to calculate boluses would be substantial. All of this is feasible in today's technological nature. A large limitation of this paper is due to the personal preferences of myself, the writer. I am the 'patient' in this context, and all the data is mine. I personally set the periods of time that the TBF operates on based on when I eat my meals, and this is in no way generalizable. Similarly, different patients have different ICRs, CF, and TargetBG. However, the default settings for these were chosen for a general patient and not based on my own.

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(Davidson et al., 2008)

[7] Schmidt et al. (2014)

A Abbreviations

BC, Bolus Calculator; CF, Correction Factor; CGM, Continuous Glucose Monitor; CHO, Carbohydrates; IBO, Insulin-on-Board; ICR, Insulin-Carb-Ratio; PS, Physiologic State; TBF, Temporal Bolus Factor